CHENQI GUO

Machine Learning | Computer Vision | Multimodal AI

Beijing, China | chenqiguo72@ncepu.edu.cn | github.com/chenqiguo | Google Scholar

PROFESSIONAL SUMMARY

Ph.D.-trained machine learning and computer vision researcher with experience spanning deep generative models, knowledge distillation, long-tailed and semi-supervised learning, multimodal learning, video evaluation, and medical AI. Currently a Lecturer at North China Electric Power University; previously a Graduate Research Associate at The Ohio State University. Co-first author of two ICCV papers, with end-to-end experience in problem formulation, algorithm design, PyTorch implementation, multi-GPU experimentation, evaluation, technical writing, and research mentoring.

EXPERIENCE

North China Electric Power University | *Lecturer, School of Control and Computer Engineering* **Jun 2023 - Present**
Beijing, China

- Lead research in machine learning and computer vision, including knowledge distillation, long-tailed/semi-supervised learning, multimodal learning, and video/model evaluation.
- Own the research lifecycle from problem definition and mathematical formulation through implementation, experiment design, model training, analysis, student mentoring, and paper writing; published first/corresponding-author work in 2025 and co-first-authored ICCV 2025.

The Ohio State University | *Graduate Research Associate, Computational Biology and Cognitive Science Lab* **Sep 2019 - May 2023**
Columbus, OH, USA

- Conducted research in deep generative models, computer vision, and medical image analysis; co-first-authored ICCV 2021 and led an automated breast-cosmesis assessment system.
- Collaborated with radiation oncologists across multiple U.S. medical institutions, translating clinical requirements into computable ML/CV problems and delivering data processing, model development, and evaluation pipelines.

SELECTED RESEARCH & ENGINEERING PROJECTS

RethinkKD: Attention, Fidelity, and Generalization in Knowledge Distillation | *Expert Systems with Applications, 2025; First & Corresponding Author* | Code

- Led the project from research conception and implementation to controlled experimentation and student mentoring; analyzed teacher attention diversity, student-teacher fidelity, mutual information, and data augmentation across ResNet/VGG/ViT models.
- Evaluated on CIFAR-100, ImageNet, and long-tailed variants; established evidence that lower student-teacher fidelity is not necessarily pathological and achieved reported Top-1 accuracy up to 53.02% on CIFAR100-imb100 and 49.68% on ImageNet-LT.

PCB: Physical Commonsense Benchmark for Generated Videos | *Ongoing Research*

- Building a benchmark and automated evaluation framework for physical plausibility in generated videos using real videos, generated videos, and controllable physics perturbations.
- Developing a pipeline from video tracking/segmentation to structured physical-state extraction, LLM-based equation discovery, physical-law fitting, deviation scoring, and validation against human annotations.

SpeLER: Spectral-Balanced Energy Propagation | *ICCV 2025; Co-first Author* | Code

- Co-developed the core algorithm and theoretical analysis for skeleton-based action recognition under class imbalance and partial labeling, including spectral-balanced label propagation and propagation-energy-based reliability estimation.
- Combined temporal and frequency-domain representations and validated the method across six datasets, achieving the leading results reported in the paper.

Fully Automatic Breast Cosmesis Assessment | *Machine Learning with Applications, 2022; First Author* | Code

- Designed and led an end-to-end CV/ML system for evaluating cosmetic outcomes after breast-conserving therapy: preprocessing -> breast/areola/nipple detection -> landmark estimation -> clinical feature extraction -> multidimensional score prediction.
- Validated on 3,762 patient images from three clinical trials with ratings from six radiation oncology experts; achieved approximately 93% / 92% / 99.8% accuracy for breast / areola / nipple landmark detection in 10-fold cross-validation.

SELECTED RESEARCH & ENGINEERING PROJECTS (CONT.)

When Do GANs Replicate? | *ICCV 2021; Co-first Author* | Code

- Studied training-data replication and memorization in BigGAN and StyleGAN2 across CelebA, Oxford Flowers, and LSUN-bedroom; found replication decreases exponentially with dataset size and complexity.
- Built a one-shot estimator that predicts replication behavior from a single observed dataset size, with a reported median prediction error of about 2.42%.

Cross-Modal Knowledge Distillation with CLIP and WordNet-Relaxed Text Embeddings | *arXiv, 2025; First Author*

- Built a multi-teacher cross-modal knowledge-distillation framework using CLIP image/text embeddings; introduced WordNet-relaxed text embeddings, hierarchical loss, and cosine regularization to reduce label leakage, with best or second-best results across six classification datasets.

Generative-Model Data Augmentation for Long-Tailed Recognition | *arXiv, 2023; First Author* | Code

- Proposed SSIM-supSubCls to characterize inter-class visual structure and found generative-augmentation gains decay approximately exponentially with this metric; built StyleGAN2-based cGAN / guided-diffusion pipelines with ResNet and MAE across four datasets.

EDUCATION

The Ohio State University | *Ph.D., Electrical and Computer Engineering; GPA 3.89/4.00* — **May 2023**, Columbus, OH, USA

Northwestern University | *M.S., Electrical Engineering; GPA 3.86/4.00* — **Jun 2019**, Evanston, IL, USA

Hebei University of Technology | *B.S., Electrical Engineering and Automation; GPA 3.91/4.00* — **Jun 2017**, Tianjin, China

SELECTED PUBLICATIONS

Bridging Class Imbalance and Partial Labeling via Spectral-Balanced Energy Propagation for Skeleton-based Action Recognition - ICCV 2025; Co-first Author

Why does Knowledge Distillation work? Rethink its attention and fidelity mechanism - Expert Systems with Applications, 2025; First & Corresponding Author

When do GANs replicate? On the choice of dataset size - ICCV 2021; Co-first Author

A fully automatic framework for evaluating cosmetic results of breast conserving therapy - Machine Learning with Applications, 2022; First Author

TECHNICAL SKILLS

Programming / Frameworks: Python, PyTorch; multi-GPU deep learning experimentation and model evaluation.
Methods: CNNs, Vision Transformers (ViT), GANs, diffusion models, CLIP / cross-modal learning, knowledge distillation, contrastive learning, long-tailed learning, semi-supervised learning, and GNNs.
Research & Collaboration: Experimental design, research code development and maintenance, technical writing, student mentoring, and cross-institution / interdisciplinary collaboration in English.